

assignment_vectors

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:54pm

Quiz Instructions

This is a challenging topic. make sure to watch the tutorials in dropbox about basic trig and about finding components of vectors (folder PDF slides in dropbox).

There is a tutorial in dropbox (see HW folder).



Question 1 11.2 pts

A velocity vector has components 36 m/s westward and 22 m/s northward. What are the magnitude and direction of this vector?

For the direction, give the angle from the + x-axis . See tutorial



42m/s@149



42m/s@31.4



Question 2 11.2 pts

The x component of vector **A** is 8.7 units, and its y component is -6.5 units. The magnitude of **A** is closest to



11 units.



8.9 units.



9.9 units.



7.9 units.



12 units.



Question 3 11.2 pts

The x component of vector **A** is 5.3 units, and its y component is -2.3 units. The angle that vector **A** makes with the +x-axis is closest to

hint: you can trace the vector in the polar graph. Then use tangent to find the angle with +x-axis

☐

250°.

☐

160°.

☐

23°.

☐

110°.

☐

340°.



Question 4 11.2 pts

The magnitude of **A** is 5.5 m, and this vector lies in the second quadrant and makes an angle of 34° with the +y-axis. The components of **A** are closest to

☐

$A_x = -4.6$ m, $A_y = 3.1$ m.

☐

$A_x = -3.1$ m, $A_y = 4.6$ m.

☐

$A_x = 3.1$ m, $A_y = -4.6$ m.

☐

$A_x = -4.6$ m, $A_y = -3.1$ m.

☐

$A_x = 4.6$ m, $A_y = -3.1$ m.



Question 5 11.2 pts

The components of vector **A** are given as follows:

$$A_x = +6.1$$

$$A_y = -8.6$$

The angle measured counterclockwise from the x-axis to vector **A**, in degrees, is closest to:

☐

125

☐

215



305



55



145



Question 6 11.2 pts

The components of vector **A** are given as follows:

$$A_x = +7.6$$

$$A_y = -6.4$$

The magnitude of **A** is closest to:



8.9



7.9



9.9



11



12



Question 7 11.2 pts

A player throws a football 50.0 m at 61.0° north of west. What is the westward component of the displacement of the football?



64.7m



24.2 m



55.0 m



0.00 m



74.0 m



Question 8 11.2 pts

A vector **A** has components $A_x = 12.0$ m and $A_y = 5.00$ m.

(a) What is the angle that vector **A** makes with the $+x$ -axis?

(b) What is the magnitude of vector **A** ?

☐

11m/s@67.4

☐

11m/s@22.6

☐

13m/s@22.6

☐

13m/s@67.4



Question 9 11.2 pts

A boy jumps with a velocity of magnitude 20.0 m/s at an angle of 25.0° above the horizontal. What is the horizontal component of the boy's velocity?

☐

9.33 m/s

☐

12.6 m/s

☐

18.1 m/s

☐

8.45 m/s

☐

15.6 m/s

Not saved

Submit Quiz